Computational Economics

Lecture: Value Functions and the Bellman Equation

Yaolang Zhong yaolang.zhong@e.u-tokyo.ac.jp
Graduate School of Economics, The University of Tokyo

Lecture Overview

- **The Value Function**
  - Policy rollouts, reward streams, and discounted returns
  - Expected return from a starting state
- **The Bellman Equation**
  - Recursive returns and policy evaluation
  - The linear system for a fixed policy
  - Optimal value and the choice of an action
- **Two Economic Models**
  - Job search: the McCall model
  - Consumption and saving with uncertain income

Recap: A Markov Decision Process

A time-homogeneous discounted **MDP** is specified by $\big(\mathcal{S}, \{\mathcal{A}(s)\}_{s\in\mathcal{S}}, P, R, \beta\big)$.

- **State space $\mathcal{S}$:** free cells $\mathcal{S} = \{\mathrm{S}, \mathrm{G}, 1, \ldots, 45\}$, including G. Walls are excluded.
- **Actions $\mathcal{A}(s)$:** up, down, left, right.
- **Transition kernel $P(s' \mid s, a)$:** probability 1 for the cell reached by action a, and 0 for all other cells. For example, $P(21 \mid \mathrm{S}, \text{down}) = 1$. A wall or boundary keeps the current cell; G is absorbing.
- **Reward $R(s, a)$:** -1 per move from outside G (including entry into G); zero at G.
- **Discount factor $\beta \in (0, 1)$:** weights future rewards.

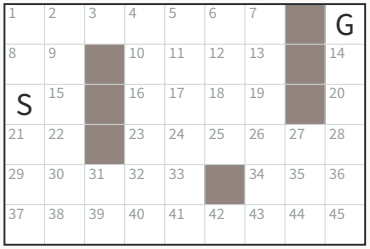


Figure 1: Maze navigation.

Recap: A Policy Chooses an Action at Each State

A **stationary deterministic policy** is a rule σ with $\sigma(s) \in \mathcal{A}(s)$: at cell s, take action $a = \sigma(s)$.

$\sigma_A = (\text{S} \to \text{down},\ \text{cell } 21 \to \text{down},\ \text{cell } 29 \to \text{right}, \ldots)$

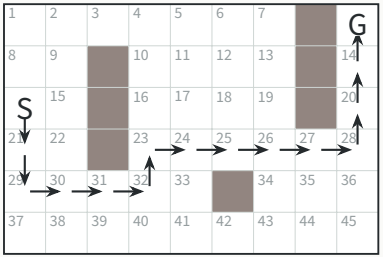


Figure 2: Policy σ_A.

$\sigma_B = (\text{S} \to \text{up},\ \text{cell } 8 \to \text{up},\ \text{cell } 1 \to \text{right}, \ldots)$

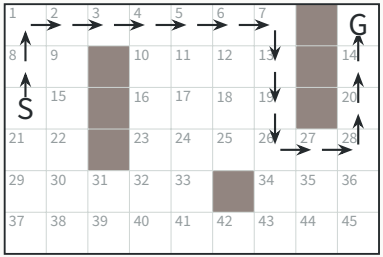


Figure 3: Policy σ_B.

Rollouts: Simulating a Policy

In reinforcement learning, a **rollout** follows a policy from a starting state, generating a **trajectory** of states, actions, and rewards.

To simulate a rollout from $s_t = s$, repeat for $\tau = 1, 2, \ldots$:

$$a_{t+\tau-1} = \sigma(s_{t+\tau-1}), \qquad s_{t+\tau} \sim P(\cdot \mid s_{t+\tau-1}, a_{t+\tau-1}).$$

Each call to $\sigma(s)$ draws an action; each call to $R(s, a)$ draws a reward. Deterministic rules always return the same outcome. Here

$$r_{t+\tau} = R(s_{t+\tau-1}, a_{t+\tau-1}).$$

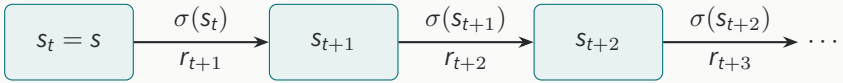


Figure 4: A rollout under policy σ.

The associated **reward stream** is $r_{t+1}, r_{t+2}, \ldots$. Rollouts can differ through actions, rewards, or state transitions.

The Return of a Rollout

The **discounted accumulated reward**, or **return**, assigns a number to the reward stream of a rollout starting at date t:

$$G_t = r_{t+1} + \beta r_{t+2} + \cdots + \beta^{\tau-1} r_{t+\tau} + \cdots = \sum_{\tau=1}^{\infty} \beta^{\tau-1} r_{t+\tau}.$$

- The first reward has weight 1; the reward τ steps ahead has weight $\beta^{\tau-1}$.
- One rollout gives one realized return. Random actions, rewards, or transitions can produce different returns under the same policy.
- To evaluate a policy from a given state, we consider the expected return over its possible rollouts.

Discounting and a Finite Return

Let $\bar{r} < \infty$ be a **uniform bound on absolute rewards**: $|r_{t+j}| \le \bar{r}$ for every $j \ge 1$.

For any rollout truncated after τ periods,

$$\left|G_t^{(\tau)}\right| = \left|\sum_{j=1}^{\tau} \beta^{j-1} r_{t+j}\right| \le \sum_{j=1}^{\tau} \beta^{j-1} |r_{t+j}|$$
$$\le \bar{r} \sum_{j=1}^{\tau} \beta^{j-1} = \frac{1-\beta^{\tau}}{1-\beta}\,\bar{r}.$$

As $\tau \to \infty$, $\beta^{\tau} \to 0$, so the infinite-horizon return satisfies

$$|G_t| \le \lim_{\tau \to \infty} \frac{1-\beta^{\tau}}{1-\beta}\,\bar{r} = \frac{\bar{r}}{1-\beta}.$$

Standing Assumptions

Unless explicitly stated otherwise, we assume:

- **Time-homogeneous dynamics.** For a given state and action, transition probabilities and the reward distribution are the same at every date:

$$P_t(s' \mid s, a) = P(s' \mid s, a), \qquad R_t(s, a) \overset{d}{=} R(s, a).$$

- **A stationary policy.** At a given state, the action distribution does not change with the date:

$$\sigma_t(s) \overset{d}{=} \sigma(s).$$

  Here $\overset{d}{=}$ means equal in distribution. A deterministic policy assigns probability one to its chosen action.

- We omit date subscripts on P, R, and σ. Over an infinite horizon, the expected return also does not depend on the starting date; we write $V^{\sigma}(s)$.

The Value Function of a Policy

The **state-value function under policy σ** is the expected return over rollouts starting at s:

$$V^{\sigma}(s) = \mathbb{E}_{\sigma}[G_t \mid s_t = s] = \mathbb{E}_{\sigma}\left[\sum_{\tau=1}^{\infty} \beta^{\tau-1} r_{t+\tau} \,\middle|\, s_t = s\right].$$

- The subscript σ in $\mathbb{E}_{\sigma}$ means that the expectation is over rollouts following this particular policy σ.
- G_t is the return of one rollout. $V^{\sigma}(s)$ averages over all possible rollouts from s under σ.
- The expectation averages over random actions $\sigma(s)$, rewards $R(s, a)$, and next states drawn from P, throughout the rollout. The starting state is held fixed.

The Return Has a Recursive Form

Along a rollout, the return from date t is:

$$G_t = r_{t+1} + \beta r_{t+2} + \beta^2 r_{t+3} + \cdots .$$

Separate the first reward from all later rewards:

$$\begin{aligned} G_t &= r_{t+1} + \beta\big(r_{t+2} + \beta r_{t+3} + \cdots\big) \\ &= r_{t+1} + \beta G_{t+1}. \end{aligned}$$

- G_{t+1} is the return from the remainder of the same rollout.
- From the next state s', the expected remaining return is

$$\mathbb{E}_\sigma[G_{t+1} \mid s_{t+1} = s'] = V^\sigma(s').$$

This lets us replace an infinite reward sequence with a one-step equation.

What Does the Expectation Average Over?

Fix the starting state $s_t = s$. There are three possible sources of randomness:

- **Action:** $\sigma(s)$ is a random action; a is one realization.
- **Reward:** the reward function $R(s, a)$ may be random; r is one realized reward.
- **Next state:** s' is drawn from the transition kernel. Conditional on reward r, use $P(s' \mid s, a, r)$.

For discrete outcomes, expand $V^\sigma(s) = \mathbb{E}_\sigma[G_t \mid s_t = s]$:

$$V^\sigma(s) = \sum_a \underbrace{\Pr\{\sigma(s) = a\}}_{\text{action}} \sum_r \underbrace{\Pr\{R(s,a) = r\}}_{\text{reward}} \sum_{s'} \underbrace{P(s' \mid s, a, r)}_{\text{next state}} \big[r + \beta V^\sigma(s')\big].$$

The mean reward and the original transition kernel are

$$\mathbb{E}[R(s,a)] = \sum_r r \Pr\{R(s,a) = r\}, \qquad P(s' \mid s, a) = \sum_r \Pr\{R(s,a) = r\} P(s' \mid s, a, r).$$

Rewards and next states may be dependent. For continuous outcomes, replace the corresponding sums by integrals.

One Starting State, Many Possible Rollouts

Restarting from the same s_0 can generate different paths. Draw again at each new state.

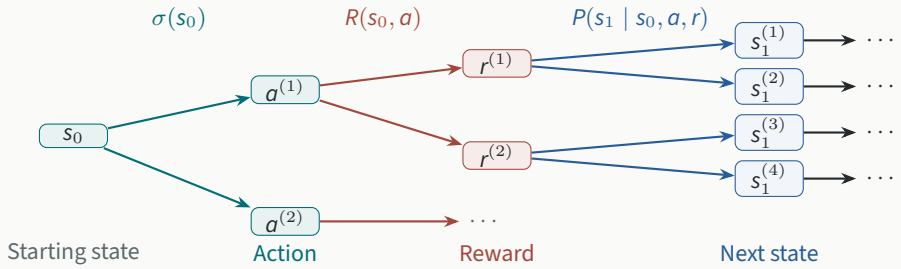


Figure 5: Branches of a rollout.

- Arrows show conditional draws; a rollout follows one branch at each step. Dots indicate further branches.
- Deterministic $\sigma(s)$ or $R(s, a)$ leaves one branch at that step.

The Bellman Equation for a Fixed Policy

For a deterministic policy $a = \sigma(s)$, average $G_t = r_{t+1} + \beta G_{t+1}$ conditional on $s_t = s$:

$$\begin{aligned} V^\sigma(s) &= \mathbb{E}_\sigma[r_{t+1} + \beta G_{t+1} \mid s_t = s] \\ &= \mathbb{E}[R(s, \sigma(s))] + \beta \mathbb{E}[V^\sigma(s_{t+1}) \mid s_t = s,\ a_t = \sigma(s)]. \end{aligned}$$

The second equality uses the law of iterated expectations and the Markov structure.

For a finite state space, the **policy Bellman equation** is

$$V^\sigma(s) = \mathbb{E}[R(s, \sigma(s))] + \beta \sum_{s' \in \mathcal{S}} P(s' \mid s, \sigma(s)) V^\sigma(s').$$

Current expected reward plus discounted expected continuation value. For a continuous state space, replace the sum by an integral with respect to $P(ds' \mid s, \sigma(s))$.

Policy Evaluation as a Linear System

For a finite state space, fix a deterministic policy σ and define

$$P^{\sigma}_{ij} = P(j \mid i, \sigma(i)), \qquad \mu^{\sigma}_{i} = \mathbb{E}[R(i, \sigma(i))].$$

- Row i of P^{σ} gives next-state probabilities when we follow σ from state i.
- Entry i of μ^{σ} is the current expected reward at that state.

Treat V^{σ} and μ^{σ} as column vectors. The policy Bellman equations become

$$V^{\sigma} = \mu^{\sigma} + \beta P^{\sigma} V^{\sigma}, \qquad (I - \beta P^{\sigma}) V^{\sigma} = \mu^{\sigma}.$$

Policy evaluation computes the value of the specified policy by solving this linear system.

From a Given Policy to the Optimal Value

For the finite-state, finite-action models below, let Σ denote the feasible stationary deterministic policies. The **optimal value function** is

$$V^*(s) = \max_{\sigma \in \Sigma} V^\sigma(s).$$

Choose the current action, then follow an optimal policy from the next state. This gives the **Bellman optimality equation**:

$$V^*(s) = \max_{a \in \mathcal{A}(s)} \left\{ \mathbb{E}[R(s,a)] + \beta \sum_{s' \in \mathcal{S}} P(s' \mid s, a) V^*(s') \right\}.$$

- The first term is expected current reward; the second is expected continuation value.
- The action is chosen before the next state is observed. Later actions can respond to the state that is realized.

The McCall Search Model: Setup

An unemployed worker observes a wage offer and chooses whether to accept it.

- **Offers:** wage $w \in \mathcal{W}$, drawn independently each period with probability $q(w)$.
- **Accept:** earn w now and every period thereafter. Employment is permanent.
- **Reject:** receive benefit b now; draw a new offer next period.
- **Objective:** maximize expected discounted income, with discount factor β.

Accept today or wait for a better offer.

The decision at date t generates income r_{t+1}.

The McCall Search Model as an MDP

The state records employment status and the current offered or accepted wage:

$$\mathcal{S} = \{(\mathrm{U}, w), (\mathrm{E}, w) : w \in \mathcal{W}\}.$$

- At (U, w), choose **accept** or **reject**. At (E, w), the only action is **work**.
- Current rewards are $R((\mathrm{U}, w), \text{accept}) = w$, $R((\mathrm{U}, w), \text{reject}) = b$, and $R((\mathrm{E}, w), \text{work}) = w$.
- The **transition kernel** is

$$\begin{aligned} P((\mathrm{U}, w') \mid (\mathrm{U}, w), \text{reject}) &= q(w'), \\ P((\mathrm{E}, w) \mid (\mathrm{U}, w), \text{accept}) &= 1, \\ P((\mathrm{E}, w) \mid (\mathrm{E}, w), \text{work}) &= 1. \end{aligned}$$

  All other transition probabilities are zero.

A policy σ specifies whether to accept each offer while unemployed.

The McCall Search Model: Recursive Values

Once employed, the state is (E, w) and there is no further choice:

$$V^{\sigma}((\mathrm{E}, w)) = w + \beta V^{\sigma}((\mathrm{E}, w)) = \frac{w}{1-\beta}.$$

For a specified acceptance policy, the **policy Bellman equation** is

$$V^{\sigma}((\mathrm{U}, w)) = \begin{cases} w + \beta V^{\sigma}((\mathrm{E}, w)), & \sigma((\mathrm{U}, w)) = \text{accept}, \\ b + \beta \sum_{w' \in \mathcal{W}} q(w') V^{\sigma}((\mathrm{U}, w')), & \sigma((\mathrm{U}, w)) = \text{reject}. \end{cases}$$

Allowing the worker to choose gives

$$V^{*}((\mathrm{U}, w)) = \max \left\{ \underbrace{\frac{w}{1-\beta}}_{\text{accept}}, \quad \underbrace{b + \beta \sum_{w' \in \mathcal{W}} q(w') V^{*}((\mathrm{U}, w'))}_{\text{reject and continue}} \right\}.$$

Consumption and Saving: Setup

A household divides its current stock between consumption and saving.

- **Choice:** from stock s, save a and consume $c = s - a$. No borrowing; storage capacity M.
- **Next period:** savings carry over one for one, plus new income Y_{t+1}, drawn independently and uniformly from $\{0, \ldots, B\}$.
- **Preferences:** utility $u(c) = c^{\alpha}$, with $0 < \alpha < 1$, and discount factor β.
- **Objective:** maximize expected discounted utility.

This extends cake eating with new income: consume now or save for periods when income is low.

Consumption and Saving as an MDP

Use whole units of goods. The state is current stock s; the action a is the amount **saved**:

$$\mathcal{S} = \{0, \dots, M + B\}, \qquad \mathcal{A}(s) = \{0, \dots, \min(s, M)\}.$$

Consumption, reward, and the next state are

$$c = s - a, \qquad R(s, a) = (s - a)^{\alpha}, \qquad s' = a + Y'.$$

The **transition kernel** is

$$P(s' \mid s, a) = \begin{cases} 1/(B+1), & a \le s' \le a + B, \\ 0, & \text{otherwise.} \end{cases}$$

- $\sigma(s)$ gives saving at stock s; consumption is $s - \sigma(s)$.
- The state space is closed: saving is at most M, and next income is at most B.

Consumption and Saving: Recursive Values

For a specified saving policy σ,

$$V^{\sigma}(s) = \big(s - \sigma(s)\big)^{\alpha} + \frac{\beta}{B+1} \sum_{y=0}^{B} V^{\sigma}\big(\sigma(s) + y\big).$$

Choosing saving optimally instead gives

$$V^{*}(s) = \max_{a \in \{0,\ldots,\min(s,M)\}} \left\{ (s-a)^{\alpha} + \frac{\beta}{B+1} \sum_{y=0}^{B} V^{*}(a+y) \right\}.$$

- Saving more lowers current utility and changes the distribution of next-period resources.
- A given saving rule σ determines P^{σ} and μ^{σ}. Solving $(I - \beta P^{\sigma})V^{\sigma} = \mu^{\sigma}$ gives its value at every stock.

References and Further Reading

1. **Richard S. Sutton and Andrew G. Barto (2018), Reinforcement Learning: An Introduction. 2nd ed. MIT Press. [Book]**
   - Chapter 3: Finite Markov Decision Processes (Sections 3.1, 3.3, 3.5 and 3.6)
   - Section 4.1: Policy Evaluation (Prediction)
   - Section 8.10: Rollout Algorithms
2. **Thomas J. Sargent and John Stachurski, QuantEcon. [Online lectures]**
   - Job Search I: The McCall Search Model — The McCall Model
   - Discrete State Dynamic Programming — Discrete DPs
   - Discrete State Dynamic Programming — Example: A Growth Model